Adaptive Retained-State Filtration for Dispersion Witnesses

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Abstract

We specify the adaptive interface required to turn low/high-dispersion estimates into a volume-uniform bad-block process while retaining the labels that carry frustration and topology.

The retained state contains dispersion class, Fréchet-center chart, center flux, boundary holonomy, orbit chart, and topology sector. Only positive-probability histories generated by the declared exposure are quantified.

Theorem 1 (Shared-uniform coupling)**.** *If every generated history h satisfies*

$$\Pr(X_{n+1} = 1 \mid h) \le q,$$

then on a common sequence of independent uniforms the entire bad field is coordinatewise dominated by independent Bernoulli(q). If $e^{\alpha} D^2 q < 1$, its rooted weighted connected sum is at most $e^{\alpha} q/(1 - e^{\alpha} D^2 q)$.

Proof. At each step compare the conditional threshold with the same uniform used for the Bernoulli variable. Induction preserves coordinatewise domination. The connected-set estimate is a geometric sum. □

The executable schema also proves the union bound $q = q_{\text{low}} + q_{\text{high}}$ and rejects incomplete history tables. The two-site law supported on 01 and 10 falsifies promotion from intersection moments alone; an erased two-state example falsifies removal of rare-entry information.

Remark 1. *The theorem is exact, but the complete nonlinear Yang–Mills history table has not been established.*